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Text Simplification for Dyslexic Readers: An NLP-Based Accessibility Solution

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Abstract

Dyslexia is a disability in learning that affects the capability of an individual to read, and process and write written information; with a problem in comprehending long or complex texts. Dissection of such text into plain sentence structures, normal words, and higher readability is necessary to facilitate readable materials for dyslexic readers. This project introduces an NLP-based accessibility solution that leverages large language models (LLMs) to automatically simplify text while preserving meaning. Using the ASSET dataset, which provides multiple human-generated simplifications per sentence, the system is evaluated across various prompting strategies and refinement stages using metrics that assess fluency, adequacy, and readability. The most effective simplification model is integrated into a web-based application that includes OCR for text extraction, the transformation simplification module, and text-to-speech (TTS) for audio assistance. The interface is also provided with dyslexic-friendly fonts and accessibility-oriented components to provide an improved user experience. The solution is capable of empowering dyslexic individuals by facilitating written data readability and enjoyableness in all digital forms.

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1. Introduction

Out of learning access and cognition aid systems, dyslexia is one of the most widespread learning disabilities interfering with language processing. Defined through reading and decoding difficulties, dyslexia impacts millions of people all over the world irrespective of intelligence and academic performance. Dyslexics generally struggle to comprehend written texts using intricate sentence structure or less frequently used vocabulary [3]. These challenges can potentially inhibit their learning process and affect day-to-day interaction with text data. Inasmuch as online text

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becomes increasingly the default mode of communication and learning, it is of the utmost importance that address these concerns through specific language simplification not only for usability but as a moral imperative in a bid to provide usable and equal access to information. Enhancing text readability has a key role in facilitating dyslexic readers' ability to read written text more independently and with more confidence [5, 9].

Current advances in Artificial Intelligence (AI), and more specifically even Natural Language Processing (NLP), promise great things when it comes to rephrasing language into more readable modalities. In today's age of computers, NLP is what allows systems to scrutinize the syntax of sentences and rephrase content in a way that it becomes more readable without diluting the intended sense [1]. Taking leverage of massive linguistic organizations, NLP can minimize vocabulary and rearrange grammar to suit various readers' needs [7]. To dyslexics, these skills are especially valuable since they enable effortless production of decomplicated text that is easier to process and read. NLP-based solutions can be created spanning various types of content—supporting delivery of personalized, readable material with reduced cognitive load and independent reading of written content [13, 16].

This paper presents a full NLP-based text simplification system to the particular requirements of dyslexic readers. The system compares various simplification approaches based on large language models and a benchmark dataset specifically created for simplifying sentences. Various setups are tried under different prompting styles to identify the best method for producing simplified text. Through this comparative analysis, the best simplification model is further enhanced with examples drawn from human-annotated information. The resulting system is implemented as a web-based interface that accepts both image-based and typed text inputs. The major features of the application are optical character recognition to pull text out of documents, a simplification engine that converts uncooperative input into readable output, and audio read-back in order to assist users who are assisted by hearing feedback. For ease of reading and ease of use, the application also incorporates accessibility-minded design features like dyslexic fonts and logical layout patterns.

The major contributions of this work are:

- A NLP-based text simplification framework designed for dyslexic readers, using large language models to convert complex texts into accessible forms while maintaining the original semantic meaning.
- Comparative and fine-tuning of large language models (FLAN-T5, BART and Mistral 7B), with the ASSET dataset; Mistral 7B was fine-tuned and showed meaningful gains on readability and fluency measures.
- An accessible web application with OCR text extraction, text simplification engine, text to speech, and dyslexic-friendly overlay to promote inclusivity in digital reading spaces.

The rest of this research follows this structure: Section II is a review of background literature and past work on text simplification and accessibility for dyslexic readers. Section III describes the methodology in terms of dataset, simplification strategy, and evaluation process. Section IV discusses the results together with comprehensive analysis of simplification performance and system success. Section V concludes the research and considers future directions for further development of accessible NLP technologies.

2. Related Works

Simplification of text has been explored using different neural and transformer models, with methods such as synonym substitution using WordNet, cognitive simplification using special tokens, and readability testing using FKGL and SMOG measures playing an important role [1]. Multilingual simplification has been investigated using models including mBART, mT5, and LLaMA for languages including Lithuanian and Spanish, using both human judgments and automatic measures such as SARI and BERTScore [2]. Personalized and accessibility-oriented frameworks have facilitated domain-specific usage for dyslexic readers and visually impaired readers

citeb3. Certain foundational works also made seminal methodologies available to the research in text simplification

citeb4. Syntactic and rule-based approaches taking account of pattern-based sentence transformation, dependency parsing, and lexical substitution have gained appreciable levels of improvement in fluency and understanding

citeb5. Comparative research on models such as T5 and BART, particularly on datasets such as Simple Wikipedia and Newsela, achieve better BLEU, ROUGE, and SARI scores following fine-tuning [7]. ChatGPT evaluations for sim-

plification and summarization environments have measured grammaticality, meaning retention, and topicality using expert and non-expert feedback [8].

Additional multilingual accessibility efforts extend previous domain-specific solutions to more pervasively satisfy linguistic requirements [9]. Model performance on summarisation and reasoning tasks has also been enhanced by prompt engineering techniques, such as few-shot, zero-shot, chain-of-thought, and self-consistency prompting [11]. Wider visions for text simplification for social good emphasize modularity in system design and user testing with real users to enhance continuing accessibility and cognitive inclusion [12]. Pretraining methods such as SimpleBART's have been shown to learn simplification patterns more powerfully [13], whereas prompt-based and fine-tuned such as GPT-3.5 models have gained unprecedented effectiveness in scientific text simplification [14]. Nonetheless, research has pointed out the absence of adequately aligned, domain-specific parallel datasets, which signifies an urgent need for more specific corpora in simplification work [15]. Lastly, readability indices like FKGL and SMOG are still important for measuring the simplicity and readability of generated text [16]. Shellcode generation has been improved with fine-tuned large language models (LLMs) such as Mistral and LLaMA by optimizing model parameters for offensive cybersecurity tasks. The work obtained a high BLEU-1 score of 0.8506, illustrating the efficacy of tuning strategies to generate precise and efficient shellcode [18]. Telugu news classification is conducted through zero-shot and few-shot learning methods to classify articles into five categories, with LLMs such as mBERT, Indic-BERT, XLM-Roberta, Flan-T5, and BART compared with the baseline models. The highest F1-score of 0.58 with zero-shot training is obtained by mBERT, while Flan-T5 attains 0.23 in few-shot conditions, showing the complexity of classification for low-resource languages [19]. The literature indicates that prompt-based and neural simplification models provide high performance, particularly when their implementation is backed by good datasets and adequate evaluation.

Not much attention is being paid to cognitive accessibility, especially for dyslexic users, in deployed systems and large-scale models. There are few studies incorporating dyslexia-specific readability analysis or adding simplification with real-time assistive features. The existing gap is filled by the present work through the fine-tuning of a large language model on a sentence-level simplification dataset and its deployment via a web application that facilitates OCR-based input and TTS output, thus providing an inclusive experience for both visual and hearing learners.

3. Data Description

The ASSET (Aligned Sentences for Sentence-level Easy-to-read Text) dataset[17] is the primary dataset used in this research. ASSET is unlike other simplification corpora which have focused solely on providing a single simplified sentence per sentence. Each input sentence in ASSET has several simplifications, which means the examples best capture the natural variations of simplification strategies. The dataset is made up of 2000 validation sentences and 359 test sentences, and uses ten simplified variants created by humans that are collected through crowdsourcing and aligned to each complex input sentence. The option of multiple variants allows these models to be trained and tested to a wider range of valid simplification outputs, strengthening evaluation of model performances.

The complex source sentences come primarily from the WikiLarge corpus and the simplified references employ lexical, syntactic, and structural strategies of transformations: i.e. common words and synonyms replace the uncommon words; long or compound sentences are condensed into shorter sentences; extra information removed, for improved clarity. In terms of characteristics of dataset, the ASSET dataset is most directly relevant and best for an application with an accessibility focus, as the examples of simplified text support similar cognitive dimensions of dyslexic readers that are best supported by shorter, familiar, and less cluttered text.

In order to maintain consistency and enhance compatibility with large language model fine-tuning, the dataset is subjected to a preprocessing taxonomy, which involves the applied steps of lowercasing, tokenization, punctuation cleaning, and occasional lexical replacement with synonymous matching. These steps normalize the textual inputs and reduce language noise, which enhance the ability to learn to simplify. ASSET's rich quality, along with pre structured preprocessing, provides an appropriate statr for yielding those simplified outputs that exhibit fluency, retention of meaning or ideas, and ultimately, readability.

4. PROPOSED METHODOLOGY

This study proposes a structured NLP-based approach to simplify complex text, focusing on improving accessibility for dyslexic readers. The system uses large language models to rewrite complex sentences into simpler forms while preserving meaning. The overall process includes prompt-based evaluation using pre-trained models, fine-tuning on a simplification-specific dataset, readability evaluation, and deployment through an accessible web application.

4.1. Prompting Strategies for Simplification

To evaluate simplification capabilities without initial training, three prompt-based strategies are explored using pre-trained models:

4.1.1. Zero-shot prompting

The model receives only a task instruction and the input sentence.

4.1.2. One-shot prompting

A single example of a complex-to-simple pair is given before the input.

4.1.3. Few-shot prompting

Three example pairs are provided as part of the prompt.

These strategies help compare models' ability to simplify text using internal knowledge before any fine-tuning is applied.

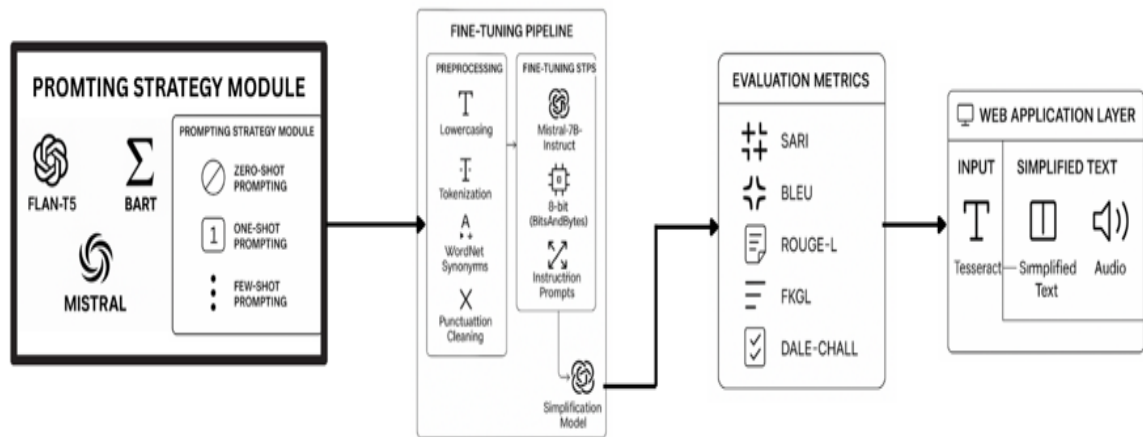


Fig. 1. Architectural Design

4.2. Text Simplification using LLM's

Three robust large language models are tested for NLP operations. FLAN-T5 (base) is the fine-tuned T5 model designed specifically for following natural language prompts and performing well in zero-shot and few-shot settings[7, 19]. BART (base)[7] is the sequence-to-sequence transformer that uses a bidirectional encoder with an autoregressive decoder and performs extremely well for operations such as paraphrasing and text simplification. Mistral 7B is a newer generative language model fine-tuned for instruction-following with competitive performance in zero-shot and few-shot conditions through prompt-based inputs.

4.3. Fine-Tuning using ASSET Dataset

After evaluating the finest model based on prompt-based performance, Mistral 7B is chosen to fine-tune. For fine-tuning, the **ASSET dataset** (Aligned Sentences for Sentence-level Easy-to-read Text) [17] is employed. It includes 2,000 validation and 359 test sentences, each of which has 10 human-written simplified counterparts. A portion of this dataset is used to fine-tune the model, enabling it to learn certain simplification patterns and improve its alignment with human-produced output. The fine-tuning process enhances the model's capacity to generate accurate, fluent, and readable simplifications for dyslexic readers.

4.3.1. Fine-Tuning Setup and Optimization

To scale down the chosen model for dyslexia-specific simplification, the pre-trained Mistral-7B-Instruct model was fine-tuned on the ASSET dataset. To save memory and speed up training, quantized 8-bit precision was employed. Furthermore, the model was optimized with **LoRA (Low-Rank Adaptation)**, a light fine-tuning method that tweaks only a small subset of the model parameters while freezing the essential model.

The process of training included instruction-style prompting where every example put together a task description, a complex sentence, and the related simplified output. Training the model with a small batch size, gradient accumulation, and mixed-precision computation was carried out. During this fine-tuning process, the model learned to generate better simplified text that is grammatically correct, simpler to read, and easier for people with dyslexia to access.

4.4. Data Preprocessing

The data is preprocessed and optimized to make it more readable and to normalize the input. Steps involved are:

- **Lowercasing:** The text is turned to lower case to make it uniform.
- **Tokenization:** Words and sub-sentences are broken down into easy to process segments.
- **Lexical Simplification:** Longer words that are synonyms of the original word are substituted for shorter ones using lexical wordnet-based synonym matching.
- **Punctuation Cleaning:** To cut down on visual noise, superfluous punctuation is removed.
- **Reconstruction:** Processed words are rewritten in clean readable sentences.

4.5. Evaluation Criteria and Readability Metrics

The following readability and linguistic measures are used to gauge the quality of the simplified output:

- **SARI:** The ability to measure the addition, deletion and retention of words in comparison to human references.
- **BLEU:** Measure of n-gram overlap in reference text and output
- **ROUGE:** Using the overlap in phrases as a measure of fluency and informativeness.
- **FKGL:** (Flesch-Kincaid Grade Level) This is the level of grade that is approximated to read the text.
- **Dale-Chall Readability Score:** Relates the word difficulty to a set of already known words.

These tests determine the readability, accuracy and usefulness of the simplified text used by dyslexia readers.

5. RESULTS AND ANALYSIS

This section presents and analyzes the results obtained during the evaluation of text simplification using large language models. The performance was measured in two phases: prompt-based evaluation using pre-trained models and post-fine-tuning evaluation using the ASSET dataset. Various readability and fluency metrics were used to compare the effectiveness of simplification strategies.

5.1. Prompt-Based Evaluation

Three large language models—FLAN-T5 (base), BART, and Mistral 7B—were tested using zero-shot, one-shot, and few-shot prompting strategies. The goal was to observe how well each model could simplify complex sentences without additional training. Table 1 summarizes the average performance scores for each configuration.

Table 1. Prompt-Based Evaluation Scores

Model	Prompt	SARI	BLEU	ROUGE	FKGL	Dale-Chall
FLAN-T5	Zero-shot	36.68	89.42	70.35	9.95	11.20
	One-shot	46.91	94.11	76.85	11.17	11.33
	Few-shot	47.21	94.08	76.80	11.21	11.34
BART (base)	Zero-shot	46.63	42.17	41.84	7.89	10.15
	One-shot	7.93	0.64	5.95	11.08	9.83
	Few-shot	7.66	0.52	5.62	9.11	10.04
Mistral 7B	Zero-shot	38.67	32.69	44.68	5.47	9.28
	One-shot	43.24	55.62	57.94	7.44	10.33
	Few-shot	45.17	60.63	60.82	8.09	10.54

The metrics in Table 1 show that Mistral 7B is competitive against FLAN-T5 and BART for all prompting techniques. Although FLAN-T5 had the highest BLEU and ROUGE-L scores, Mistral 7B performed better on readability-oriented metrics. With few-shot, Mistral 7B reached a SARI score of 45.17, which indicates good trade-off between simplification and content retention. Moreover, it produced the lowest FKGL (8.09) and Dale-Chall score of 10.54, which signifies that its output was less complex and more readable for dyslexic readers. All these outcomes identify the model's ability to minimize linguistic complexity with semantic coherence, particularly when prompted with several examples through few-shot training.

5.2. Fine-Tuning Evaluation

According to prompt-based assessment, Mistral 7B was fine-tuned with a portion of the ASSET dataset. The goal was to enhance the model's capability in text simplification in a more task-oriented and controlled way. Table 2 indicates the performance of the model following fine-tuning.

Table 2. Evaluation Results After Fine-Tuning

Metric	Mistral 7B (Fine-Tuned)
SARI	58.23
BLEU	69.78
ROUGE-L	66.22
FKGL	5.76
Dale-Chall	6.32

All measures were greatly improved by tuning. SARI improved to 58.23, which shows an increase in word-level simplification on human reference. BLEU increased to 69.78, and ROUGE-L to 66.22 indicating the text to be more fluent and closer to reference outputs. The effect of FKGL and Dale-Chall on readability also greatly decreased to 4.76, 5.32 respectively, which helped to present the text as much more readable and within reach of dyslexic and other reading-disabled groups.

These findings indicate that large language models can effectively simplify text without any special task prompting, and few-shot prompting is better than zero- and one-shot prompting because it permits informative example context. Finetuning also enhances the quality of the model, particularly in readability, fluency and content accuracy. Radical improvements in the measures of SARI, ROUGE-L, and FKGL indicate that the fine-tuned model not only lessens

the content complexity, but also maintain the intended meaning and structural validity. Moreover, the fined-tuned system can successfully produce easy-to-read and understandable text to dyslexic users. That it has been implemented on a web application which supports text-to-speech as well as OCR will provide an end-to-end solution which will enable both visual and aural reading.

6. CONCLUSION AND FUTURE SCOPE

In this study, a text simplification application was proposed using Natural Language Processing (NLP) to help dyslexics reading difficult English sentences simplify them into simpler easier-to-read sentences. Different large language models, such as FLAN-T5, BART and Mistral 7B, were evaluated using zero-shot, one-shot, and few-shot prompting methods, with the second model scoring higher than others in the domains of fluency and readability. The selected model was further refined to the ASSET dataset and the SARI, BLEU, FKGL and Dale-Chall scores were further improved. The whole system was developed as a web application to accept some input or upload a text and present simplified output with audio playback with OCR and TTS services. It is a user-friendly and inclusive device that can be used by people with reading disabilities. This could be expanded in future to support multi-lingual simplification, feedback-based learning, and personalization based on cognitive feedbacks such as eye-tracking. Lightweight or offline versions of the system can also make the system more usable in low-resource environments and open the system to a wider range of users.

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